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Modelling the abundance of mosquito vectors versus flooding dynamics

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Abstract

An approach allowing the estimate calculation for the local abundance of mosquito vectors as a function of the variations of environmental conditions of their breeding habitats is developed. To characterize the mosquito reproductive habitats and associated vectorial dynamics, we have introduced the concepts and derived expressions of the (i) vectorial production capacity of the breeding habitat, which gives the number of vectors the breeding site is capable to produce per day, (ii) the susceptibility or availability function of the breeding habitat which measures the degree of overlap in the course of time between oviposition and emerging surfaces, and (iii) the time-dependent vectorial reproduction number, resulting from the combination of (i) and (ii), which characterizes the growth dynamics of the vector population. As an illustrative application, the method analysis is employed to characterize the potential of mosquito production of two ground pools known as likely sources of vectors of Rift Valley Fever.

Keywords: Mathematical modelling; Arbovirus vectors; Ground pool; Discharge dynamics

1. Introduction

Arboviruses are mainly transmitted to vertebrates through infected bites of arthropods during their blood meals (Edman, 2000). Arbovirus infections affect humans and animals by often causing febrile illnesses and sometimes severe neural troubles or haemorrhagic fevers citepeld. To cite a few, the Rift Valley fever, West Nile fever, Dengue or Yellow fever are all arbovirus infections for which principal vectors are mosquito species of the genera *Aedes* and/or *Culex* (Eldridge et al., 2000; Fontenille et al., 1998; Pratt and Moore, 1993; Linthicum et al., 1990). It has been understood that emergence and/or re-emergence of arbovirus infections result partly from the tremendous increase in the number of vectors and of the widespread and proliferation of their reproduction habitats that depend on environmental variables. The underlying rationale is that environmental conditions control the formation and multiplication of vector breeding habitats, and the dynamics of vectors controls in turn the virus transmission and

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circulation within vertebrate populations. Given these relationships, the ecology of arbovirus infections requires studies and understanding of both, the dynamics of reproduction habitats and that of vector population within an established breeding nest.

Indeed, the vector mosquito species oviposit in specific aquatic surfaces or habitats (Pratt and Moore, 1993; Bicout, 2001) . The number of new emerging imago depends on the aquatic or humid surface that receives eggs, allows eggs development to hatching, and constitutes a suitable nest for forthcoming transformations to emergence of imago. These surfaces are not identical for all mosquito species since depending on each respective bio-ecology. As examples of breeding sites, the *Aedes* species lay their eggs on depressions on damp soils or above mean high water while the *Culex* species directly oviposit batch of eggs on water surfaces (Crans and McNelley, 1998; Pratt and Moore, 1993). In addition, each type of associated breeding sites is characterized according to physico-chemical factors in relation with surrounding soil and vegetation. That is to say that the impact of flooding water dynamics on reproduction surfaces is not identical for all mosquito species. Furthermore, the transmission potential of arboviral diseases (defined as the reciprocal of the number of female mosquito vectors/host required such that the basic reproduction number of the disease is equal at least to one) is related to the vector density (Rogers and Randolph, 2000; Patz et al., 1998a). Therefore, assessing the dynamics of vector populations within a given area can be considered as a direct sensitive indicator of favorable conditions for the potential risk of viral transmission.

For this purpose, we have constructed a theoretical framework allowing the estimate calculation of the abundance of mosquito vectors as a function of the variations of environmental conditions of their breeding habitats. The abundance of vectors, after appropriate scaling, is the quantity which enters into the basic reproduction number for the disease spread. Our approach to tackle the problem is based on physical mechanism of the formation of breeding habitats and on bio-ecology of mosquitoes. We restrict our study to mosquito of genera *Aedes* and *Culex* that, like for the Rift Valley Fever for instance, are found to play complementary roles of vector-reservoir and vector-amplifier, respectively, in the transmission of virus (Bicout and Sabatier, 2004; Fontenille et al., 1998; Linthicum et al., 1990). Two dynamical models are developed to address these issues: first model describes the respiration dynamics of a ground pool, considered as the main mosquito breeding habitat (Fontenille et al., 1998; Linthicum et al., 1990). The outputs of this latter model are next used in a second model for the description of the population dynamics of vectors. For a concrete illustrative application, the method analysis as developed uses field data of limnigraphic records from two ground pools (Barkedji, Senegal) to characterize the potential of these pools of producing mosquitoes.

2. Ground pool discharge dynamics

Our goal in this section is to construct a simplest minimal model that captures main features of the emptying pond dynamics without any knowledge of pond characteristics that require long and heavy studies. We consider in this description the fundamental processes of the discharge dynamics to characterize the pond with corresponding parameters. The kind of ponds we will be dealing with are formed of two regions (Fig. 1): a clogged region, of height h_c, situated at the bottom of the pond and above which there is an unclogged region. Clogging originates from silt clayey accumulations of material caused by erosion mechanisms (aeolian, hydric, ...) (Desconnets et al., 1997). Three key processes control the dynamics of the pond: the water inputs from rainfall and runoff, the infiltration of water in the soil and the evaporation of water in the air. Water inputs are function of the weather and the surrounding hydro-system, the infiltration a characteristic of the soil forming the pond and the evaporation a characteristic of environmental conditions. Infiltration of water in the unclogged region is fast depending on the water load while it appears to be relatively slow and almost constant in the clogged region (Desconnets et al., 1997).

Let h be the filling level of the pond at time t, the emptying dynamics of the pond between two water inputs can be described by the following differential equation:

$$\frac{\mathrm{d}h}{\mathrm{d}t} = -v_{\mathrm{r}} - \kappa(h)\,\mathrm{e}^{-\omega t}\,[h - h_{\mathrm{c}}] + \Delta h(t), \tag{1}$$

where the first term at the right hand side of the sign equal is the slow discharge due to residual evaporation (i.e., evaporation plus infiltration in the clogged region)

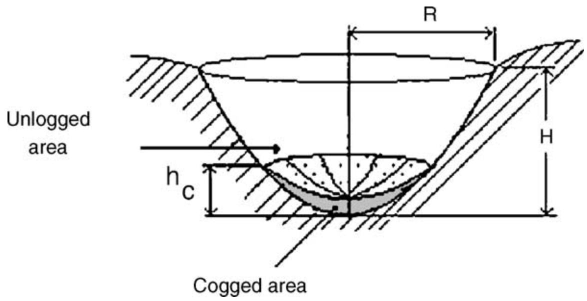

Fig. 1. Schematic description of a typical pond.

with constant speed v_r, the second term represents the fast infiltration in the unclogged region with the time dependent leaking rate $\kappa(h)\ \exp\{-\omega t\}$ such that $\kappa(h) = \kappa$ when $h > h_c$ and zero for $h \leq h_c$. The clogging time $1/\omega$ is the time scale beyond which the infiltration in the unclogged region becomes identical to that in the clogged one. The last term represents time-dependent water inputs of level $\Delta h(t)$. Eq. (1) has to be solved with the initial condition, $h(t = 0) = h_0 \geq 0$ to obtain $h(t)$ for $t \geq 0$. This model is intended to be used to obtain draining parameters.

As an illustration, this simple model for the emptying dynamics as outlined above is used to describe limnigraphic row data from two ground pools, Mous1 and Mous2, located in Barkedji area (15°20′N, 14°80′W), in Senegal. Mous1 is an elliptical pond of about 800 m long by 100 m large with scattered light vegetation while Mous2, almost circular, is about one third smaller than Mous1. These ponds, situated in the same geographic region, are subjected to identical weather conditions, and specially, to identical rainfall pattern. A χ^2 adjustment between the theory and data is employed to determine the parameters of the model. The result is depicted on Fig. 2.

Fig. 2 clearly shows that limnigraphic records, although different for the two ponds, are very well described by the theoretical model. Inspection of emptying parameters indicates that although these ponds share the same weather conditions, their size and the physical characteristics of their soils are rather different in consistency with simple observation. Mean values of the draining parameters are reported in Table 1 for information. It appears that Mous1 is

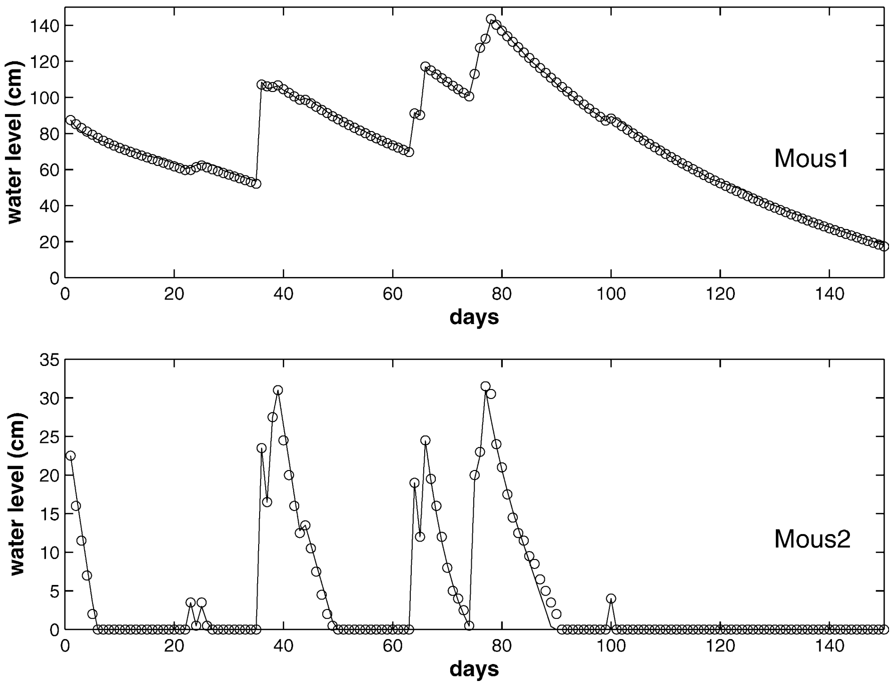

Fig. 2. Discharge dynamics of two ponds in Barkedji (Senegal) during the 2001 rainy season. Circles represent data and solid lines the theory. Highest water levels are $H = 143.5$ cm for Mous1 and $H = 31$ cm for Mous2.

Table 1
Mean values of draining parameters with Δh as an input field data

Definition	Symbol	Unit	Mous1	Mous2
Clogging level	h_c	cm	81.8	1.3
Residual evaporation	v_r	cm/days	1.0	3.8
Leakage	κ	1/days	0.029	0.137
Clogging time	$1/\omega$	days	∞	∞
Input level	Δh	cm	–	–

deeper and larger than Mous2, because of larger clogged region in Mous1, but the residual evaporation in Mous2 is faster than in Mous1, suggesting a faster infiltration in clogged region in Mous2. Furthermore, the rate of leaking in the unclogged region is larger in Mous2 than in Mous1. These findings emphasize that the simple discharge dynamics model developed above is able to capture main features of the pond activity in response to environmental variations. As shown in Fig. 2, Mous1 shows less activity and is a semi-temporary pond at the time scale of the rainy season while the more active Mous2 is a temporary pond which even dries out during the season.

The main goal of such a modelling is to determine the draining parameters of a pond in order to be able to further anticipate on values of $h(t)$ provided that $\Delta h(t)$ is known in other respects. Although the model may become parameter rich as the emptying parameters are different for each rain shower, such a study allows one to outline trends in variations of the soil characteristics for a better modelling. Now we know the variations in time of the water level $h(t)$, which turns out to be a key element in the formation of mosquito oviposition sites, we may turn to the study of the dynamics of vector populations.

3. Vectorial dynamics

Natural life cycle of mosquitoes involves four developmental stages: three aquatic stages including egg, larva, and pupa, plus an aerial one which is adult or imago (Pratt and Moore, 1993). Female mosquitoes (vectors) lay their eggs on specific humid surfaces S called oviposition sites. The form of surfaces and their physico-chemical characteristics are crucial for favorable conditions of forming a site for potential occupation of eggs, larval and pupa habitats. Fig. 3 displays a schematic illustration of oviposition sites (or reproductive sites) for two species of mosquitoes involved in transmission of arboviruses.

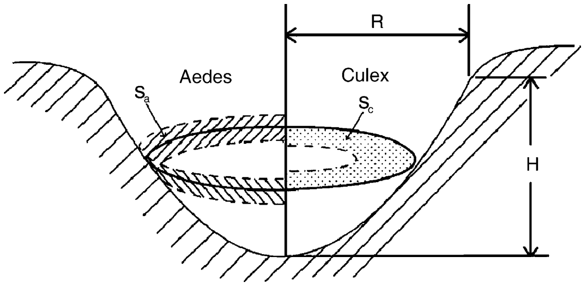

Fig. 3. Schematic illustration of oviposition sites for *Aedes* (S_a) and *Culex* (S_c) mosquitoes.

In this study, we are interested to assess and characterize the abundance of the vectors, i.e., the total number of individual vectors per unity of a reference (like a trap or a given space), produced from a given mosquito reproductive site. As this number of vectors is a random variable depending of the intrinsic dynamics of mosquito life cycle and of the collecting method, we denote by $M(t)$ the expected abundance of adult vector mosquitoes at time t determined from a given capture procedure. In what follows, we will deal with the situation where there are no sources of mosquitoes, emigration and immigration processes of mosquitoes do not take place, and crowding effects are negligible. Starting with a initial population $M(t=0)=M_0$ of adult mosquitoes in a reproductive site, the dynamics of the abundance of vectors can be described by the delayed differential equation:

$$\frac{\mathrm{d}M}{\mathrm{d}t} = -\alpha M + \sigma\chi(t,T)M(t-T), \qquad (2)$$

where $1/\alpha$ is the vector lifetime and T denotes the developmental period, i.e. the elapsed time during which a newly laid egg undergoes a maturation and transformation from egg stage to emergence of an adult (Table 2). The time-dependent production rate $\sigma\chi(t,T)$ of new mosquito vectors is the product of the vectorial production capacity, σ, of the breeding site and of the susceptibility or availability function, $\chi(t,T)$, of the reproduction habitat. The production rate of mosquitoes is derived as follows.

Table 2
Parameters involved in the vectorial dynamics

Definition	Symbol	Unit
Life expectancy	$1/\alpha$	days
Developmental period	T	days
Imago emergence probability	β	$0 \le \beta \le 1$
Female fecundity	λ	eggs/female/days
Surface capacity of the habitat	S_{m}	cm^2
Pre-imago survival probability	$\phi(T)$	$0 \le \phi(T) \le 1$
Effective level of emergence	$h(t\|t-T)$	cm
Highest water level	H	cm
Geometric exponent of the habitat	ν	$\nu \ge 0$

Let S_{m} be the surface of the oviposition site and λf the number of eggs per unit of time a female mosquito can deposit in such a site. For practical uses, S_{m} can be defined as the maximum surface range of the site, λ is the number of eggs per unit of time laid by a female mosquito and the scaling factor $f = f(\lambda, S_{\mathrm{m}})$ rescales the total surface occupied by eggs with respect to the size of the site S_{m}. At earlier time $t - T$, a fraction $S(t-T)/S_{\mathrm{m}}$ of the oviposition site of surface $S(t-T)$ (such that $0 \le S(t) \le S_{\mathrm{m}}$ for all t) is available for mosquito laying. In this oviposition site, a female mosquito may lay $\lambda\, fS(t-T)/S_{\mathrm{m}}$ eggs out of which a proportion $\phi(T)$ survives, undergoes hatching to larval and transformation to pupa stages up to the time of emergence t. Simultaneously to the maturation and transformation phases, the habitat surface undergoes changes from $S(t-T)$ to $S(t|t-T)$ where $S(t|t-T)$, that incorporates bio-ecology of mosquitoes and physico-chemical characteristics of the habitat, represents the smallest surface during the developmental period that still contains stages susceptible of leading to emergence of adults. Thus, at time t, just a fraction (or probability) $S(t|t-T)/S_{\mathrm{m}}$ of surviving pupa $\phi(T)\lambda\, fS(t-T)/S_{\mathrm{m}}$ have a chance β of giving rise to adults. As a result, the production rate of new vectors from a female mosquito is given by $\sigma\chi(t,T) = \beta S(t|t-T)\phi(T)\lambda\, fS(t-T)/S_{\mathrm{m}}^2$. This rate is specific of mosquito species, the associated breeding habitat and depend upon environmental conditions. In addition, all parameters involved in the production rate cannot be experimentally determined one by one. In this respect, all static terms are lumped together into the vectorial production capacity:

$$\sigma = \beta\lambda f\phi(T), \tag{3}$$

and all dynamic terms are described be the availability function:

$$\chi(t,T) = \frac{S(t|t-T)S(t-T)}{S_{\mathrm{m}}^2}. \tag{4}$$

By definition, σ is the expected number of mosquito vectors per unit of time that a breeding site is able to produce and the dynamic term $\chi(t,T)$ (with $0 \le \chi(t,T) \le 1$), which measures the degree of overlap between oviposition and emerging surfaces, can be regarded as the survival probability of the potential that a breeding site has of producing mosquitoes in the course of time. It is therefore natural to define from this the vectorial basic reproduction number associated to a breeding site as:

$$R_0(t,T) = \frac{\sigma\chi(t,T)}{\alpha} = r_0\,\chi(t,T), \tag{5}$$

which represents the expected number of vector's offspring produced out of the breeding site that a female mosquito can generate during its lifespan $1/\alpha$. With this definition, $\chi(t,T)$ can also be reinterpreted as the reduced vectorial reproduction number. Thus, by definition of the reproductive number, two conditions $\chi(t,T) \ne 0$ and $r_0 > 1/\chi(t,T)$ must be satisfied simultaneously for the increasing of the population of mosquitoes. This defines the production threshold $\sigma_{\mathrm{s}} = \alpha/\chi(t,T)$ above which the vector population can grow and below which it stutter to die. The $R_0(t,T)$ is specific of mosquito species, their associated breeding habitats, patterns and time variations of environmental conditions. In other words, different mosquito species breeding in the same site will be likely related to different $R_0(t,T)$. As such, $R_0(t,T)$ can be used as an indicator of suitable conditions of vector–environment coupling for the potential risk of viral transmission.

If we denote by $\psi(t)$ the distribution of mosquito lifetime (that, in Eq. (2), is an exponential, $\alpha\,\mathrm{e}^{-\alpha t}$), the abundance of the population of vectors at any time can be obtained from:

$$M(t) = \sum_{n=0}^{\infty} M_n(t) \quad \text{with}$$

$$M_n(t) = \int_T^t \psi(t-t')R_0(t',T)M_{n-1}(t'-T)\,\mathrm{d}t', \tag{6}$$

where $M_n(t)$ is the n-th generation population of mosquitoes defined for $t \geq nT$ with $M_0(t) = M_0 \int_t^\infty \psi(t')\,dt'$. Eq. (6) is quite general and admits a simple explanation: the n-th generation population of mosquitoes at time t is equal to the $(n-1)$-th generation population at earlier times $t' - T$ times the number of emerging mosquitoes (reproduction number) at t' averaged over the emergence times t' by the distribution of survival lifetime from the onset time T to t.

Another quantity of interest is the first generation population, which in the case of exponential distribution for $\psi(t)$, is

$$R(t) = \frac{M_1(t)}{M_0} = \alpha\,e^{-\alpha(t-T)} \int_T^t R_0(t', T)\,dt'. \qquad (7)$$

This function is informative on the expected number of vector offspring produced after a single round of a mosquito life cycle. Like $R_0(t, T)$, the $R(t)$ is also specific of mosquito species, their associated breeding habitats, patterns and time variations of environmental conditions. When the conditions of breeding site do not vary much in time, one may approximate $R_0(t, T)$ by a constant, i.e., $R_0(t, T) \simeq R_0$. In this case, $M_n(t)$ is a shifted Poisson distribution given by:

$$M_n(t) = M_0\,\frac{[\alpha(t-nT)\,R_0]^n}{n!}\,e^{-\alpha(t-nT)}\ ;\ t \geq nT\,. \qquad (8)$$

This function increases like $(t - nT)^n$ for short times, attains its maximum at $t = nT + 1/(n\alpha)$, and exponentially decreases to zero at long times.

Now, we specialize to *Aedes* and *Culex* mosquito genera which are known to be involved in the transmission of number of arboviruses ((Bicout and Sabatier, 2004; Fontenille et al., 1998; Linthicum et al., 1990)). As shown in Fig. 3 and mentioned above, *Culex* mosquitoes lay their eggs in raft directly on the water surface while *Aedes* species lay their eggs on moist soils above mean high water. In the case where the reproductive habitat of mosquitoes is a ground pool as in Fig. 3, the surfaces of the oviposition sites is directly related to the water level following the scaling $S \sim h^\nu$, where ν (increases with the slope of the pond) is the geometric exponent which characterizes the slope of the oviposition site associated to a mosquito species. It follows that the expression of σ is still given by Eq. (3), whereas that of the susceptibility function (or the reduced vectorial reproduction number) becomes:

$$\chi(t, T) = \left[\frac{h(t|t-T)h(t-T)}{H^2}\right]^\nu, \qquad (9)$$

where H is the maximum range of the pool water level. For a parabolic shaped ground pool, the geometric exponent of the oviposition sites is,

$$Aedes: \quad \nu = \tfrac{1}{2} \quad \text{and} \quad Culex: \ \nu = 1. \qquad (10)$$

In addition to the geometry, we need to define the effective level $h(t|t-T)$ for emergence of imago, which incorporates the bio-ecology of mosquitoes. This can be done, for instance, by taking into account that *Aedes* eggs show resistance and survive to stressed dry conditions while *Culex* eggs do not and are destroyed in the same conditions. In mathematical terms, this can be expressed as:

$$\begin{cases} Aedes: \quad h(t|t-T) = \min[h(t), h(t-T)], \\ Culex: \quad h(t|t-T) = \min[h(t')], \\ \qquad \text{for } t - T \leq t' \leq t. \end{cases} \qquad (11)$$

These characteristics already highlight the bio-ecological differences between the two species of mosquitoes and, consequently, anticipate on differences in the dynamics of their abundance.

As an illustrative application, we employ the above analysis to characterize the ground pools Mous1 and Mous2 (known as foci of mosquitoes) in Barkedji (Senegal) in terms of susceptibility of producing mosquitoes and of reproductive number of vectors in the course of time. To this end, we use the water level $h(t)$ extracted from data analyzes of limnigraphic records as described in the previous section to compute the functions $\chi(t, T)$ and $R(t)$ as functions of time. For the sake of comparison and to emphasize on effects of the kind of breeding habitats, bio-ecology of mosquitoes and environmental variations, we assume in the calculations that (i) the onset of mosquito emergence, (ii) the developmental period, $T = 10$ days, and (iii) the vectorial basic reproduction number, $r_0 = 200$, are identical for the two mosquito genera *Aedes* and *Culex* in the two ponds under study. Note that, in contrast to the reality, a very large value of r_0 is used here for the purpose of illustration. The results of calculations are displayed in Figs. 4 and 5.

On one hand, Fig. 4 shows that, for Mous1, which does not dry out during the rainy season, both the sus-

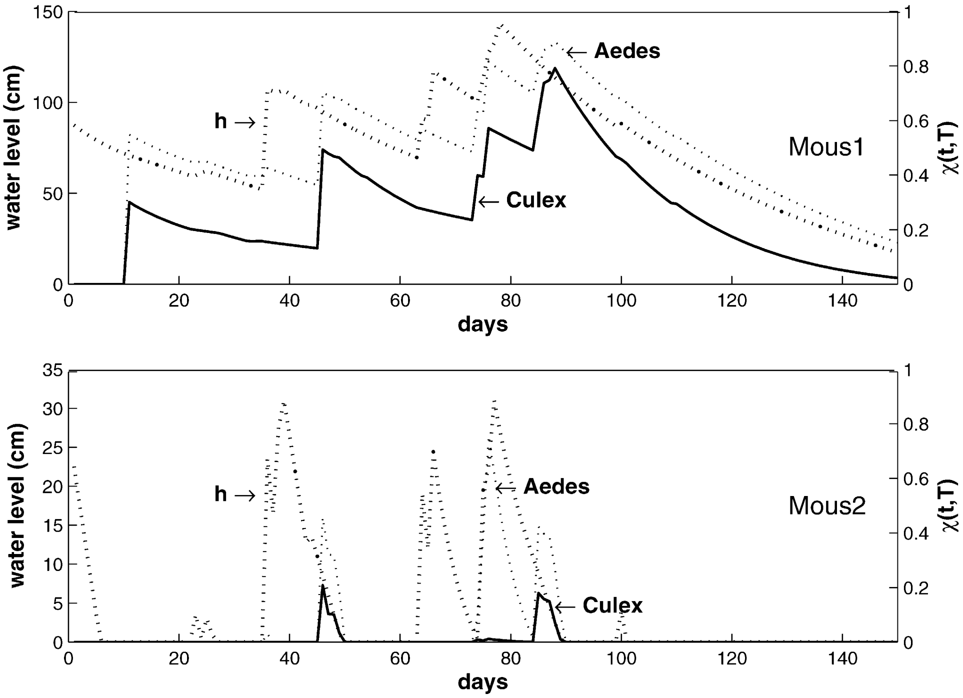

Fig. 4. Water level and theoretical susceptibility or availability function in Eq. (9) of mosquito production for two ponds in Barkedji (Senegal) during the 2001 rainy season. Dots represent data of the pond water level, dashed and solid lines availability of *Aedes* ($\nu = 1/2$) and *Culex* ($\nu = 1$) species, respectively, with $T = 10$ days and values of H given in Fig. 2.

ceptibility of *Aedes* and *Culex* production exhibits similar behavior following the water level variations but with a forward shift in time corresponding to the developmental period. The difference noticed in numerical values of the $\chi(t, T)$ is due to geometric exponents of breeding habitats; smaller ν leads to larger $\chi(t, T)$. In contrast, for the more active pond Mous2, which does empty during the rainy season, the $\chi(t, T)$ for the two mosquito genera are very different both in their behaviors versus water level and in their numerical values. In this case, the divergence in the behavior of susceptibility functions (or reduced vectorial reproduction number) is in part due to the slope of breeding sites and mainly due to the bio-ecology of mosquitoes as described in Eq. (11).

On the other hand, Fig. 5 shows that the first generation population of mosquitoes is more than one fold greater in Mous1 than in Mous2. In Mous1, $R(t)$ exhibits the time behavior like in Eq. (8) for $n = 1$, and therefore, attains its maximum at $t \simeq T + 1/\alpha = 30$ days in consistency with a permanent pond undergoing no major variations of breeding conditions. In addition, $R(t)$ for *Aedes* increases when lowering the geometric exponent of breeding sites, i.e., in decreasing the slope of the pond. The behavior of $R(t)$ as a function of time is similar for *Aedes* and *Culex* in Mous1 while it is different and exhibits peaks in Mous2. These general trends in mosquito (first generation) abundances versus time are very consistent with field data (Fontenille et al., 1998) collected in the same region, although a direct comparison is not possible because of the lack of limnigraphic records for these data. Under the conditions (i) to (iii) defined above, these findings allow one to conclude that the pond Mous1 appears to be more susceptible of producing mosquitoes than Mous2 under the same environmental conditions, and that the population of *Aedes* is substantially more abundant than that of *Culex* for a given ground pool. However, it is worthwhile to note that the present picture may no longer be the same when the tenets (i) to (iii) are not identical for the two mosquito species as it happens in natural conditions. However, once these conditions are given,

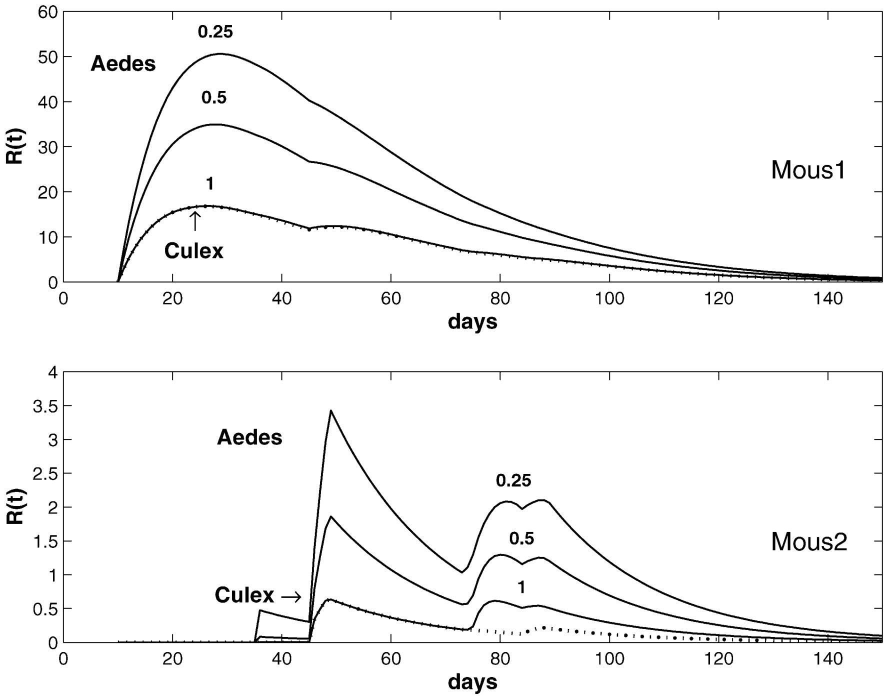

Fig. 5. First generation vectorial reproduction number (in Eq. (7)) of *Aedes* (solid line) and *Culex* (dot line) species for two ponds in Barkedji (Senegal) during the 2001 rainy season. Quoted numbers denote geometric exponents for *Aedes* while $\nu = 1$ for *Culex*. The other parameters are: $1/\alpha = 20$ days, $T = 10$ days, and $r_0 = 200$ with H given in Fig. 2.

this analysis may help to anticipate the dynamics of mosquito population.

4. Concluding remarks

Mapping vector habitats and habitat quality in relation with mosquito bio-ecology, weather and environmental factors are problematic of the landscape epidemiology approach (Meade et al., 1988; Pavlovsky, 1966). In this context, we have developed a theoretical framework allowing the estimate calculation of the abundance of vectors as a function of the variations of environmental condition of the breeding habitats. To characterize the reproductive habitats and associated vectorial dynamics, we have introduce the concepts and derive expressions of the vectorial production capacity of the breeding habitat, which gives the number of vectors that a breeding site is capable to produce per day, the susceptibility or availability function which measures the degree of overlap in the course of time between oviposition and emerging surfaces, and the time-dependent vectorial reproduction number which characterize the growth dynamics of the population of vectors. We note in passing that similar concepts like the vectorial capacity, as previously introduced in malaria epidemiology by Garrett-Jones (1964), is now largely used in literature for vector-borne diseases.

Our aim in developing the method analysis outlined in this study is to ultimately improve the assessment and enhance predictive modelling of potential risk of circulation and transmission of pathogens. This method can also be incorporated and combined with new techniques of remote sensing and geographical information systems for identifying and mapping landscapes related to diseases transmission risk (Shaman et al., 2002; Patz et al., 1998b; Linthicum et al., 1987, 1990). Finally, this work can be embellished in number of directions by including, for instance, surrounding vegetation or land cover, phyto-chemical and physico-chemical factors in breeding habitat characteristics, or survival strategies of mosquitoes like diapause and hibernation which are epidemiologically important respectively in the persistence of viruses in breeding habitats and in the amplification of virus circulation (Bicout et al., 2002). Such a work is underway.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at 10.1016/j.ecolmodel.2004.06.044.

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